

A Full Higher-Smoothness Answer for L_p Embeddings into $c_0(\Gamma)$

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Status

This packet gives a likely valid full answer to the higher-smoothness question asked after Corollary 28 in Hajek–Johannis–Schlumprecht [1]. The source corollary states that if X is a closed subspace of $L_p(\mu)$, $1 < p < \infty$, with $\text{dens } X < \omega_\omega$, then X admits a bi-Lipschitz embedding into some $c_0(\Gamma)$ whose coordinate functions are C^1 smooth. The authors then ask whether, in the case $p \geq 2$, the coordinate functions can have higher smoothness.

The answer is yes: for every $p \geq 2$ in that L_p -branch, one can choose the coordinates C^2 . More generally, the coordinates can be chosen C^m for every integer $m < p$, and for every finite m when p is an even integer.

Source Question

The source passage is on page 11 of arXiv:2401.00831.

the following corollary (note that any Lipschitz function from a subset of a metric space can be extended to the whole space with the same Lipschitz constant):

Corollary 28. *Let X be a subspace of $L_p(\mu)$ for some measure μ and $1 < p < \infty$, resp. of some super-reflexive Banach lattice with a (long) unconditional basis or a weak unit, with $\text{dens } X < \omega_\omega$. Then there is a bi-Lipschitz embedding $\Phi: X \rightarrow c_0(\Gamma)$ such that the component functions $f_\gamma \circ \Phi$ are C^1 -smooth, where f_γ are the canonical coordinate functionals on $c_0(\Gamma)$.*

We do not know whether in the corollary above in the case $p \geq 2$ the component functions can be of higher smoothness. Invoking [HJ] Theorems 7.79 and 7.86] we obtain a corollary on approximation of Lipschitz mappings:

Corollary 29. *Let X be a subspace of $L_p(\mu)$ for some measure μ and $1 < p < \infty$, resp. of some super-reflexive Banach lattice with a (long) unconditional basis or*

In text form, after the C^1 -coordinate embedding corollary, the source asks:

We do not know whether in the corollary above in the case $p \geq 2$ the component functions can be of higher smoothness.

The parameter p occurs in the $L_p(\mu)$ alternative of the corollary, so the sentence asks about the L_p -case for $p \geq 2$.

Background: Property $(*^m)$

Definition 1. Let $m \in \mathbb{N}$. A Banach space X has property $(*^m)$ if there is a constant $C_0 \geq 1$ such that for every Lipschitz function $f : X \rightarrow \mathbb{R}$ and every $\varepsilon > 0$, there is a C^m -smooth Lipschitz function $g : X \rightarrow \mathbb{R}$ satisfying

$$|f(x) - g(x)| < \varepsilon \quad (x \in X), \quad \text{Lip}(g) \leq C_0 \text{Lip}(f).$$

Fry and Keener prove that a WCG Banach space with a C^m smooth norm has property $(*^m)$ [3, Theorem 4]. Their theorem is stated for C^p smoothness; here we apply it with $p = m$.

Proof Intuition

The source already does the hard metric work: density below ω_ω gives a bi-Lipschitz embedding into $c_0(\Gamma)$. The question is only whether the scalar coordinates can be smoothed to order greater than one.

The smoothing mechanism is independent of order. The Hajek–Johanis coordinate-smoothing theorem used by the source replaces Lipschitz scalar coordinate tests by smooth Lipschitz approximants, choosing the approximation error below the separation margins that make the map bi-Lipschitz and below the thresholds that keep the image c_0 -valued. If the scalar approximants are C^m , the resulting coordinates are C^m , with no new metric bookkeeping.

For subspaces of L_p , $p \geq 2$, the needed C^2 scalar approximation property follows from the C^2 smoothness of the L_p norm and Fry–Keener. This already answers the source question. The same reasoning gives the sharper C^m range allowed by the L_p norm.

Order- m Coordinate Smoothing

Lemma 1 (Order- m coordinate smoothing). Let $m \in \mathbb{N}$. Suppose X has property $(*^m)$. If X admits a bi-Lipschitz embedding into $c_0(\Gamma)$, then X admits a bi-Lipschitz embedding into some $c_0(\Gamma')$ whose coordinate functions are C^m smooth.

Proof. This is the order- m form of the Hajek–Johanis coordinate-smoothing theorem invoked in the source proof of the C^1 -coordinate corollary [2, Theorem 7.63]. We record why the differentiability order is a formal parameter.

The construction starts from Lipschitz scalar coordinate tests associated to a given $c_0(\Gamma)$ -embedding. These tests are placed in thresholded families so that, for each $x \in X$, the resulting coordinate vector lies in c_0 , and for each pair x, y some coordinate detects $\|x - y\|$ up to a fixed multiplicative constant. The final step replaces each scalar Lipschitz test by a smooth Lipschitz approximant. The replacement only has to satisfy two quantitative conditions: its Lipschitz constant is controlled, and its uniform error is smaller than the margins used in the separation and threshold estimates.

For C^1 coordinates, this replacement uses property $(*^1)$. If $(*^m)$ is available, the same replacement gives C^m -smooth approximants with the same quantitative bounds. The rest of the argument uses only those bounds, not first derivatives or any feature special to $m = 1$. Hence the final $c_0(\Gamma')$ -valued map is still bi-Lipschitz and all of its coordinate functions are C^m . $\square$

Main Theorem

Theorem 1. *Let X be a closed subspace of $L_p(\mu)$, where $2 \leq p < \infty$, and suppose that $\text{dens } X < \omega_\omega$. Then X admits a bi-Lipschitz embedding into some $c_0(\Gamma')$ whose coordinate functions are C^2 smooth.*

More generally, if $m \in \mathbb{N}$ and $m < p$, the coordinates may be chosen C^m . If p is an even integer, the coordinates may be chosen C^m for every finite m .

Proof. By the source paper’s L_p embedding theorem, equivalently the L_p -case feeding Corollary 28, the assumption $\text{dens } X < \omega_\omega$ gives a bi-Lipschitz embedding

$$X \longrightarrow c_0(\Gamma)$$

for some set Γ [1, Corollary 26].

Since $1 < p < \infty$, $L_p(\mu)$ is reflexive, and therefore each closed subspace X is reflexive and in particular WCG. If $m < p$, the usual L_p -norm is C^m smooth away from the origin; if p is an even integer, the same standard smooth-norm argument gives all finite orders. Restricting this norm to X gives a C^m smooth norm on X in the stated range.

Fry–Keener [3, Theorem 4] then gives property $(\ast^m)$ for X . Applying Lemma 1 to the source embedding yields a $c_0(\Gamma')$ -embedding with C^m -smooth coordinate functions.

For the full answer to the source question, take $m = 2$. If $p > 2$, then $m = 2 < p$. If $p = 2$, p is an even integer, so the even-integer case again gives C^2 coordinates. Thus every $p \geq 2$ in the source question admits coordinates of higher smoothness than C^1 . $\square$

Verification Notes

The proof has no computational component. The external inputs are:

- the source paper’s L_p -subspace bi-Lipschitz embedding into $c_0(\Gamma)$ for density below ω_ω ;
- Hajek–Johannis coordinate smoothing, used with differentiability order m instead of order 1;
- Fry–Keener’s Lipschitz C^m approximation theorem on WCG spaces with C^m smooth norms;
- the classical smoothness range of L_p norms.

Limitations

This packet answers the sentence in the source involving $p \geq 2$, hence the L_p -branch of Corollary 28. It does not claim a corresponding higher-smoothness theorem for the separate super-reflexive Banach-lattice alternative unless those spaces are independently known to have property $(\ast^m)$. For non-even p , it also does not claim coordinate smoothness in orders $m \geq p$.

Novelty Check

The local run indexes were searched for arXiv:2401.00831, the source title, “higher smoothness”, “component functions”, C^m/C^2 coordinate phrases, Fry–Keener, and $c_0(\Gamma)$ embedding terminology. The only hits were the earlier conditional packet and the intermediate partial packet from this run. A bounded web search on June 29, 2026 for the source title together with “higher smoothness”, “component functions”, and $p \geq 2$ found the source arXiv page and no later explicit answer.

Human Review Recommendation

Review Lemma 1 first. Once the order- m reading of the Hajek–Johanis coordinate-smoothing theorem is accepted, the rest of the proof is a direct combination of the source embedding theorem, Fry–Keener, and standard L_p -smoothness.

References

- [1] P. Hajek, M. Johanis, and T. Schlumprecht, *Remarks on the point character of Banach spaces and non-linear embeddings into $c_0(\Gamma)$* , arXiv:2401.00831, 2024.
- [2] P. Hajek and M. Johanis, *Smooth analysis in Banach spaces*, De Gruyter Series in Nonlinear Analysis and Applications, vol. 19, De Gruyter, Berlin, 2014.
- [3] R. Fry and L. Keener, *Approximation of Lipschitz functions by Lipschitz, C^p smooth functions on weakly compactly generated Banach spaces*, arXiv:0907.0241, 2009.